Mentl: proof-carrying control software for the physical world

Hard Tech Week 2026 · Nucleus × Startup State open call · [[name]] · [[email]] · github.com/ampactor-labs/mentl · github.com/ampactor-labs/sonido

THE PROBLEM

Software that steers physical things (inverters, flywheels, flight controls) is tested, reviewed and hoped on: no mainstream language lets a team state "this loop never reaches the network, never allocates, never commands past this bound" and have the compiler prove it. Coding agents now write that software, and the grid is where the gap bites first: every new inverter is a software product under UL 1741 SB, and Utah's data centers are the load.

WHAT EXISTS TODAY

Mentl is a programming language and self-hosting compiler with an algebraic effect system, compiled to WebAssembly. A negated effect such as `!Network` is a claim proven through every call or the program does not build; every fact carries its reason; a learning rule is derived from the chain it learns on; a paused computation is saved by copying bytes. The pinned compiler reproduces itself byte for byte. [[years]] of work, open source (Apache and MIT). Beside it, sonido: a Rust DSP framework whose kernels run unchanged in plugins, a browser editor and a guitar pedal.

BUILT THIS WEEK (SINCE MONDAY 4:00 PM)

- HUM: the grid's frequency heard live from a room's wiring through a guitar pickup, read to the millihertz three independent ways; a Torus Station modeled from Torus's published figures answering in two bands; a real 2019 grid disturbance replayed from open data.
- The controller written in Mentl: proven unable to reach the network, touch the filesystem, allocate mid-decision or command either machine past its rating; every decision explained in words; a twin with one price lookup added, refused with the path named.
- The learner (a frequency tracker whose gradient the compiler derives), the page, the pedal as a second ear, an agent's proposal refused by the proof gate.

WHO NEEDS IT, IN UTAH

- **Torus** (South Salt Lake): the Station hears every cycle; the program beneath the dispatcher has never been provable. The Torus judge saw this demo at the same hackathon.
- **Hill AFB, 309th Software Engineering Group**: Utah's largest employer of software developers, verifying flight software by review; a proof is a deliverable.
- **Every Utah company putting agents on production code**: a gate that answers PROVEN or REFUSED before anything an agent wrote runs.

MILESTONES

Day 30	The proven controller on inverter-class hardware on the bench. SAM.gov and SBIR.gov registered; the company formed [[date / status]].
Day 60	One Utah site: an ear on a Station-class inverter, the dispatch explained to its operator. The SBIR Phase I draft with Nucleus Grow (AFWERX open topic or a DOE Office of Electricity topic): an assurance tool that proves properties of a flight-software module.
Day 90	Ninety days of decisions on the page; the proposal filed; the first paid pilot signed.

THE $10,000

$4,000	inverter-class development hardware and a GPS clock for the second ear
$3,000	ninety days of the proof gate hosted for the pilot and the first agent-running teams
$3,000	SBIR preparation, registration and travel

PEOPLE AND RESOURCES I NEED

- Nucleus Grow for the SBIR lane and the Phase I draft.
- An introduction at Torus engineering (via Nicole Witmyer) and at the 309th (via 47G).
- A co-founder with power-electronics depth. Connecting people is what this program does; this is my request.

Business: paid pilots with asset operators; a hosted proof gate for teams running coding agents; the SBIR lane for software assurance. The demo runs from the repository (`demos/grid-rush-hour`); the proofs run with `mentl check`. [[one line: who you are, in your words]]